ITA0610-Machine Learning

LAB EXPERIMENTS

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Lab 2: Candidate-Elimination Algorithm

Aim

To implement the **Candidate-Elimination algorithm** to find the version space by determining the most specific and most general hypotheses consistent with the training examples.

Algorithm

1. Initialize the **Specific hypothesis (S)** using the first positive training example.
2. Initialize the **General hypothesis (G)** with the most general values ?.
3. Process each training example.
4. For a positive example, generalize the specific hypothesis where required.
5. For a negative example, specialize the general hypothesis to exclude that example.
6. Remove inconsistent hypotheses.
7. Display the final specific and general hypotheses.

Program

```
# Lab 2: Candidate-Elimination Algorithm


import numpy as np


data = np.array([
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
])


# Initialize specific hypothesis
S = list(data[0][:-1])
```

```
# Initialize general hypothesis
G = [['?' for _ in range(len(S))]]

for row in data:
    x = row[:-1]
    label = row[-1]

    if label == 'Yes':
        for i in range(len(S)):
            if S[i] != x[i]:
                S[i] = '?'

    else:
        new_G = []
        for hypothesis in G:
            for i in range(len(S)):
                if S[i] != '?' and S[i] != x[i]:
                    h = ['?' for _ in range(len(S))]
                    h[i] = S[i]
                    new_G.append(h)

        if new_G:
            G = new_G

print("Final Specific Hypothesis:")
print(S)

print("\nFinal General Hypothesis:")
for hypothesis in G:
    print(hypothesis)
```

Sample Output

Final Specific Hypothesis:

['Sunny', 'Warm', '?', 'Strong', '?', '?']

Final General Hypothesis:

['Sunny', '?', '?', '?', '?', '?']

['?', 'Warm', '?', '?', '?', '?']

['?', '?', '?', 'Strong', '?', '?']

Result

Thus, the **Candidate-Elimination algorithm was successfully implemented**, and the final specific and general hypotheses were obtained from the given training examples.